BCA (Sem-2) – Discrete Structures (202) Core-IV

2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

Compulsory

1. Choose the correct alternative for each of the following questions : 2×10=20

   (i) Let P and Q be statements, then P<->Q is logically equivalent to______

a) $P \leftrightarrow \sim Q$

b) $\sim P \leftrightarrow Q$

c) $\sim P \leftrightarrow \sim Q$

d) None of the mentioned

(ii) The compound statement $A \vee \sim (A \wedge B)$.

F T
F T
T T
T F

a) True

b) False

(iii) What is the dual of $(A \wedge B) \vee (C \wedge D)$?

a) $(A \vee B) \vee (C \vee D)$

b) $(A \vee B) \wedge (C \vee D)$

c) $(A \vee B) \vee (C \wedge D)$

d) $(A \wedge B) \vee (C \vee D)$

(iv) A compound proposition that is always ______is called a tautology.

a) True

b) False

(v) A compound proposition that is always ______is called a tautology.

a) True

b) False

(vi) Let Q(x) be the statement "x < 5." What is the truth value of the quantification $\forall X Q(x)$, having domains as real numbers.

a) True

b) False

(vii) A ____is an ordered collection of objects.

a) Relation

b) Function

c) Set

d) Proposition

(viii) The set O of odd positive integers less than 10 can be expressed by _____

a) {1, 2, 3}

b) {1, 3, 5, 7, 9}

c) {1, 2, 5, 9}

d) {1, 5, 7, 9, 11}

(ix) What is the Cartesian product of A= {1,2} and B = {a, b} ?

a) {(1, a), (1, b), (2, a), (b, b)}

b) {(1, 1), (2, 2), (a, ai, (b, b)}

c) {(1, a), (2, a), (1, b), (2, b)}

d) {(1, 1), (a, a), (2, a), (1, b)}

(x) The set of positive integers is ____

a) Infinite

b) Finite

c) Subset

d) Empty

Section–B

Short Answer type Questions

2. Answer any **four** questions of the following :

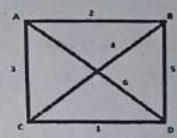
5×4=20

(a) Show that $f(x)=x^2+2x+1$ is $O(x^2)$

(b) Evaluate the sum $\sum_{k=1}^{0}(5k^2+8k+1)$

(c) Use the laws of logic to show that the following expression is a tautology $((p \to q) \land p) \to q$

(d) What is spanning tree? Find the minimum spanning of the given weighted graph



(e) A farmer buys 3 cows, 2 pigs and 4 hens from a man who has 6 cows, 5 pigs and 8 hens. Find the number m of choices that farmer has ?

(f) Use the substitution method. Solve the recurrence relation

$T(n)=T(n-1)+n$

T(1)=1

(g) Solve the recurrence relation $F_n = 10F_{n-1} - 25F_{n-2}$ where $F_0 = 3$ and $F_1 = 17$

Section–C

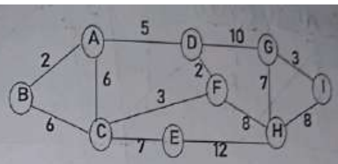
Long Answer type Questions

3. Answer any **two** questions of the following : 15×2=30

(a) Solve the recurrence relation $F_n = 5n - 1 - 6F_{n-2}$ where $F_0 = 1$ and $F_1 = 4$.

(b) Use the laws of logic to show that $p \leftrightarrow q$ and $(p \to q) \wedge (q \to p)$ are equivalent. Find the minimum no of students in a class to be sure that 4 if them were born in same month.

(c) Write prism's algorithm. Find the minimum cost spanning tree of the graph given below using prism algorithm



(d) Solve the recurrence relation $F_n = 3F_{n-1} + 10F_{n-2} + 4.5^n$ where $F_0 = 4$ and $F_1 = 3$.

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